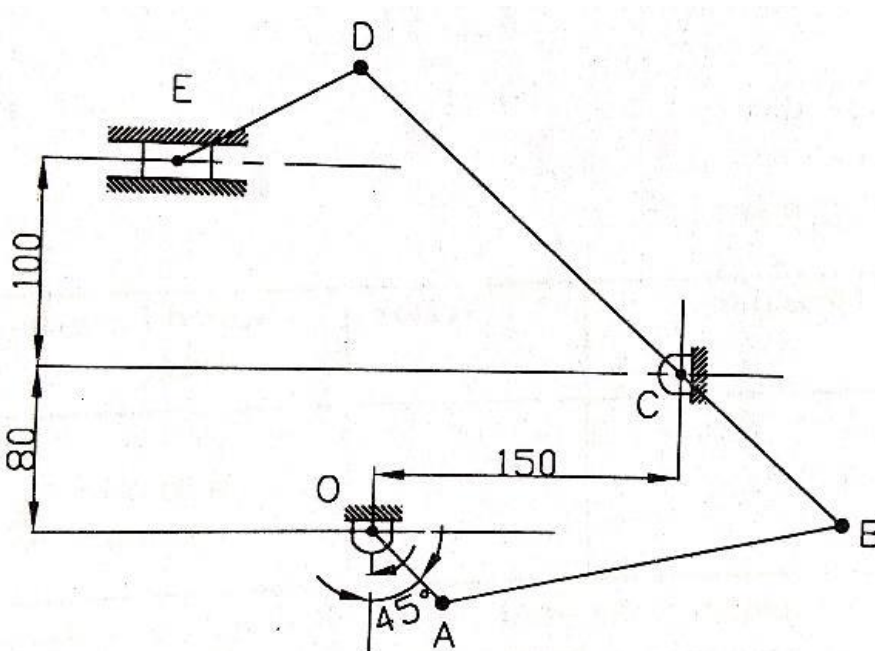
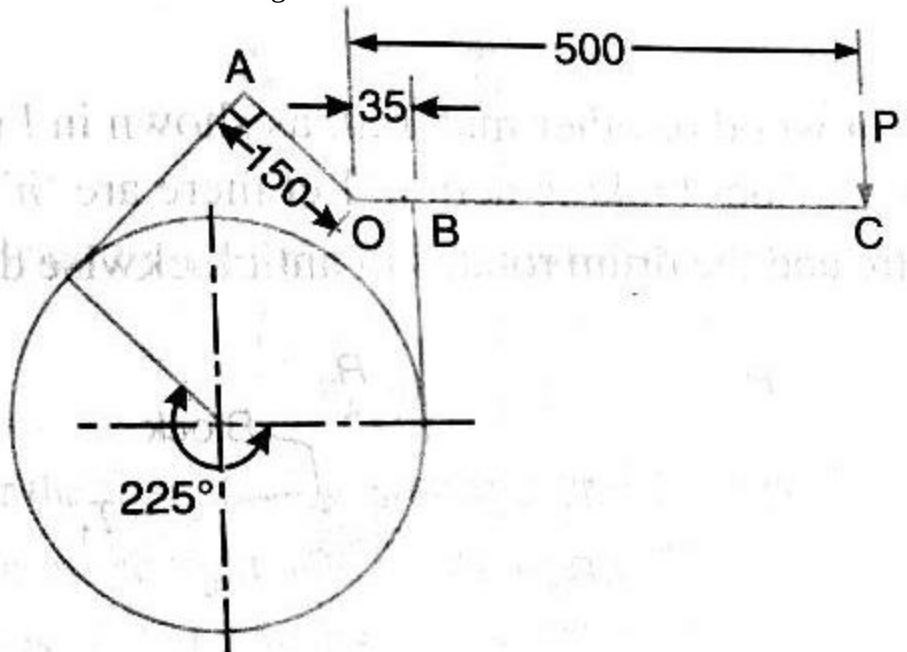


	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch :Mech.Engg Semester :IV Subject Code & Name: BTMEC402 Theory of Machines I Max Marks: 60 Date: Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO) Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)	
A)	Fig. 1 shows part of an opposed piston engine mechanism. The velocity of the piston E for the given instant is 780 mm/s. the crank OA rotates at a uniform speed in a clockwise direction and makes an angle of 45° to vertical as shown in Fig. 1. Draw a velocity diagram and determine the speed of the crank in rpm. OA= 50mm, AB= 200 mm, BC= 110 mm, CD= 220 mm and DE= 100 mm.  Fig No 01	6
B)	Explain with sketches different types of constrained motions.	6
C)	How Kinematic pairs are classified ? Explain with neat Diagram	6

Q.2	Solve Any Two of the following. (This is just a sample instruction)		
A)	Derive the expression for a total torque acting on a truncated conical pivot bearing considering uniform wear,		6
B)	A shaft has a number of collars integral with it. The external diameter of the collar is 400 mm and the shaft diameter is 250 mm. if the intensity of the pressure is 0.35 N/mm ² and the coefficient of friction is 0.05. Estimate power absorbed when the shaft runs at 105 rpm carrying a load of 150 kN and a number of collars required.		6
C)	Write a short note on Friction types with suitable example.		6
Q. 3	Solve Any One of the following. (This is just a sample instruction)		
A)	A differential band brake as shown in Fig. 2 has an angle of contact of 225°. The band has a compressed woven lining and bears against a cast iron drum of 350 mm diameter. The brake is to sustain a torque of 350 Nm and the coefficient of friction between the band and drum is 0.3. Find the necessary force 'P' for the clockwise rotation of the drum and the value of 'OA' for the brake to be self-locking when the drum rotates in clockwise.  All dimensions in mm. Fig No 02		6
B)	Differentiate between absorption type dynamometers and transmission type dynamometers		6
Q.4	Solve Any Two of the following. (This is just a sample instruction)		
A)	How cams are classified? Explain in details with neat sketches.		6
B)			6

	Draw the profile of a cam operating a roller reciprocating follower having a lift of 40 mm. the roller diameter is 20 mm. the minimum radius of the cam is 30 mm. the cam raises the follower with simple harmonic motion for 110° of its rotation followed by a period of dwell for 80°.the follower descends for the next 120° rotation of the cam follower with uniform acceleration and deceleration followed by a dwell period.		
C)	Explain with figures any three types of followers.		6
Q. 5	Solve Any One of the following. <i>(This is just a sample instruction)</i>		
A)	A four-cylinder vertical engine has cranks 150 mm long. The planes of rotation of the first, second and fourth cranks are 400 mm and 200 mm respectively from the third crank and their reciprocating masses are 50 kg, 60 kg and 50 kg respectively. Find the mass of the reciprocating parts for the third cylinder and the relative angular positions of the cranks in order that the engine may be in complete primary balance.		6
B)	Explain the method of balancing of several masses in the same planes		6
	*** End ***		

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